Status	Finished
Started	Wednesday, 21 October 2026, 12:27 PM
Completed	Wednesday, 21 October 2026, 12:33 PM
Duration	5 mins 36 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Quick Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Binary Search
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Extraction
c. Selection
d. Insertion

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Decreases the efficiency of algorithms
c. Makes data harder to manage
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. Each part is searched for a specific element
c. The parts are ignored
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Arranging data in a specific order
- c. Removing duplicates from a list
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **7**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **8**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is difficult to implement
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Merge sort
c. Heap sort
d. Quick sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It sorts data out of the box efficiently
c. It requires external libraries
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It allows for quicker identification of duplicates
c. It complicates the analysis process
d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **15**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Bubbling up the largest element to its correct position with each pass
c. Being the most efficient sorting algorithm
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass